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Kidney Supportive Care in Advanced Chronic Kidney Disease

Sara N. Davison

CHAPTER OUTLINE

KEY POINTS

- “Kidney supportive care” (KSC) is replacing the term “kidney palliative care.” KSC is an approach that aims to improve the health-related quality of life (HRQOL) for people and their families for whom kidney disease, either directly or indirectly, substantially impacts their well-being, treatment options, or access to care. This is accomplished through the prevention and relief of suffering by means of early identification and impeccable assessment and treatment of pain and other problems, physical, psychosocial, and spiritual. KSC should be provided throughout the continuum of illness, regardless of life expectancy. It can be provided together with therapies intended to prolong life, such as kidney replacement therapy, and requires culturally sensitive shared decision making to prioritize the components of medical care most important to the patient.
- Dialysis treatment may address some symptoms, such as fatigue, anorexia, nausea, and vomiting, especially for more robust individuals with less comorbidity, but it may do little to address symptom burden and HRQOL in more frail patients or those with multimorbidity. When taken as a whole, symptom burden is similar in predialysis kidney failure patients, those treated with conservative kidney management, and those on chronic dialysis.
- The Edmonton Symptom Assessment System: Renal is a simple tool to screen for 12 commonly experienced symptoms in CKD using 0 to 10 visual analog scales.
- The aim of symptom management is to ameliorate symptoms that are burdensome and adversely impact the patient’s HRQOL as it is not always necessary or possible to resolve them completely. This usually requires that symptoms be treated to ≤3/10.
- Gabapentin can be used to effectively treat neuropathic pain, restless legs, uremic pruritus, and insomnia if used carefully in low doses and titrated slowly to effect.
- Dialysis is unlikely to benefit patients with significant preexisting functional or cognitive impairment and high levels of comorbidity. These patients are often better cared for with conservative kidney management and careful attention to the principles of KSC.

Despite advances in predialysis care and dialysis technology, people with advanced chronic kidney disease (CKD) continue to have a shortened life expectancy and poor outcomes including physical, emotional, and spiritual suffering, as well as low health-related quality of life (HRQOL).[1] The majority of patients die in acute care facilities, with aggressive care plans in place[2] and without accessing palliative care services.[3,4] Current end-of-life care practices are not consistent with patients' preferences.[5,6]

Supportive care is central to the provision of patient-centered care for patients with chronic diseases. Many countries are placing increased emphasis on the provision of supportive and end-of-life care by "generalist" and community providers as a component of usual care. Therefore kidney supportive care (KSC) is increasingly being recognized as a core clinical competency for the care of patients with advanced CKD. Unfortunately, nephrologists are often not trained adequately to address multifactorial suffering and many of the end-of-life challenges inherent in the care of their patients.[7] Consequently, patients with advanced CKD experience significant unmet care needs.[5,8,9] Training in KSC for clinicians treating people with CKD is an urgent priority. This chapter aims to serve as an introduction to KSC and provide some of the foundations for understanding and delivering the key components of KSC.

DEFINING KIDNEY SUPPORTIVE CARE

Palliative medicine has undergone many developments over the past several decades and has extended the range of services, the timing of services within the illness trajectory, and the eligible patient groups such that palliative care is no longer limited to terminal care focused primarily on those dying with cancer. Unfortunately, despite the evolution of palliative care over the past several decades, the belief that "palliative care" and "terminal care" are synonymous, such that only patients at the end of life are appropriate for palliative care, remains prevalent among patients, families, and health care providers.[10,11] This is not appropriate for people with advanced CKD who frequently have high palliative care needs for years before death. For this reason, the term "kidney supportive care" is now preferred.[1,12-14] KSC is defined as "an approach that aims to improve the HRQOL for people and their families, for whom kidney disease, either directly or indirectly, substantially impacts their well-being, treatment options, or access to care, through the prevention and relief of suffering by means of early identification and impeccable assessment and treatment of pain and other problems, physical, psychosocial, and spiritual.[12] This definition is rooted in the World Health Organization (WHO) definition for palliative care.[15] KSC is not restricted to people who withdraw from dialysis or who receive conservative kidney management (CKM). End-of-life care (sometimes referred to as "terminal care" or "hospice care") is under the umbrella of KSC but is typically limited to the care of people who are believed to be within months of death (Fig. 61.1). Ideally, KSC should be started early so that issues of suffering are addressed as they present. As illness progresses, the need for KSC is likely to increase and will ultimately progress into terminal, end-of-life care.

KEY COMPONENTS OF KIDNEY SUPPORTIVE CARE

KSC requires the culturally sensitive shared decision making to 1. prioritize the components of medical care most important to the individual and 2. ensure those priorities guide clinical decisions.[12,16-20] The core components of KSC include symptom management, crisis planning, advance care planning, spiritual care, end-of-life care considerations, and integration with community services. Patient engagement has highlighted many of these issues as top priorities for people with advanced CKD.[21] A comprehensive approach to KSC must also integrate a way to identify patients most likely

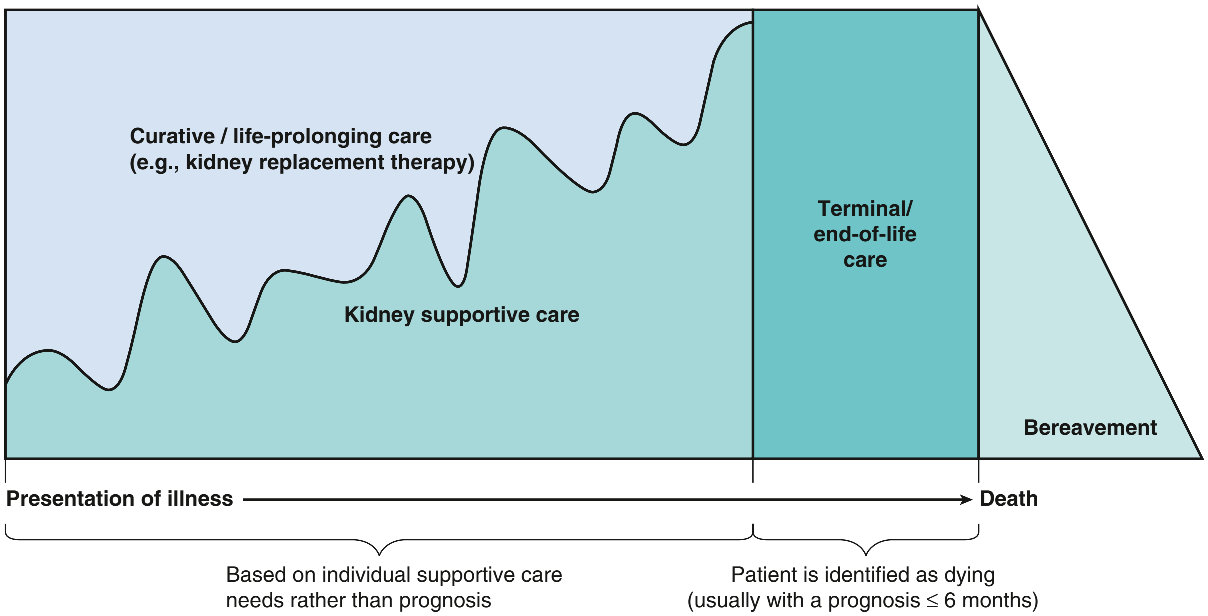

Fig. 61.1 Conceptual framework for kidney supportive care.

to benefit from supportive care interventions (Fig. 61.2). Each of these components of KSC is discussed within the context of CKD.

CULTURALLY COMPETENT SHARED DECISION MAKING

Culturally competent shared decision making is the foundation of KSC and must be incorporated into all components from assessment and management of suffering, sharing prognosis, and advance care planning to delivery of end-of-life care and bereavement support (see Fig. 61.2). To relieve suffering and provide care aligned to the preferences and goals of the individual patient, care teams must understand and incorporate the patient's needs and perspective and adapt the care plan to facilitate integration of the patient's lifestyle including his or her family and social community. This means allowing the components of medical care deemed most important to the patient to be prioritized.[16-20] This may require greater emphasis being placed on managing bothersome symptoms rather than maximizing long-term health outcomes such as survival (e.g., balancing the management of dizziness and fatigue with optimal blood pressure control or serum chemistry). As disease progresses, patients' goals of care often shift to focus almost exclusively on HRQOL rather than survival, with a strong emphasis on emotional, social, and family support.[5,22] The ethical imperatives of shared decision are discussed further in Chapter 80.

Culture has enormous potential to influence patients' beliefs around life and death, healing and suffering, as well as the physician–patient relationship.[23] Cultural sensitivity is particularly important in delivering end-of-life care, and cultural variations in end-of-life preferences have long been noted.[24] In multicultural societies, very different, and often divergent, value systems may be at work, and health care providers will need to be cognizant of these cultural influences. Although a more detailed exploration of how various cultural perspectives may influence the provision of KSC can be found elsewhere,[23] several high-level issues are discussed here.

There are tremendous cross-cultural differences in preferences for decision making. Western cultures tend to promote "patient autonomy," where the patient is viewed as the best person to make health care decisions. However, many cultures are characterized by strong communal and social bonds where social relationships, rather than individualism, provide the basis for moral judgments.[23] From this perspective, an insistence on self-determination erodes the value placed on personal interconnectedness and challenges the assumption that the patient should make his or her own medical decisions in isolation from their community. In practice, this might mean that the family or community receives and discloses information, even when the patient is competent.

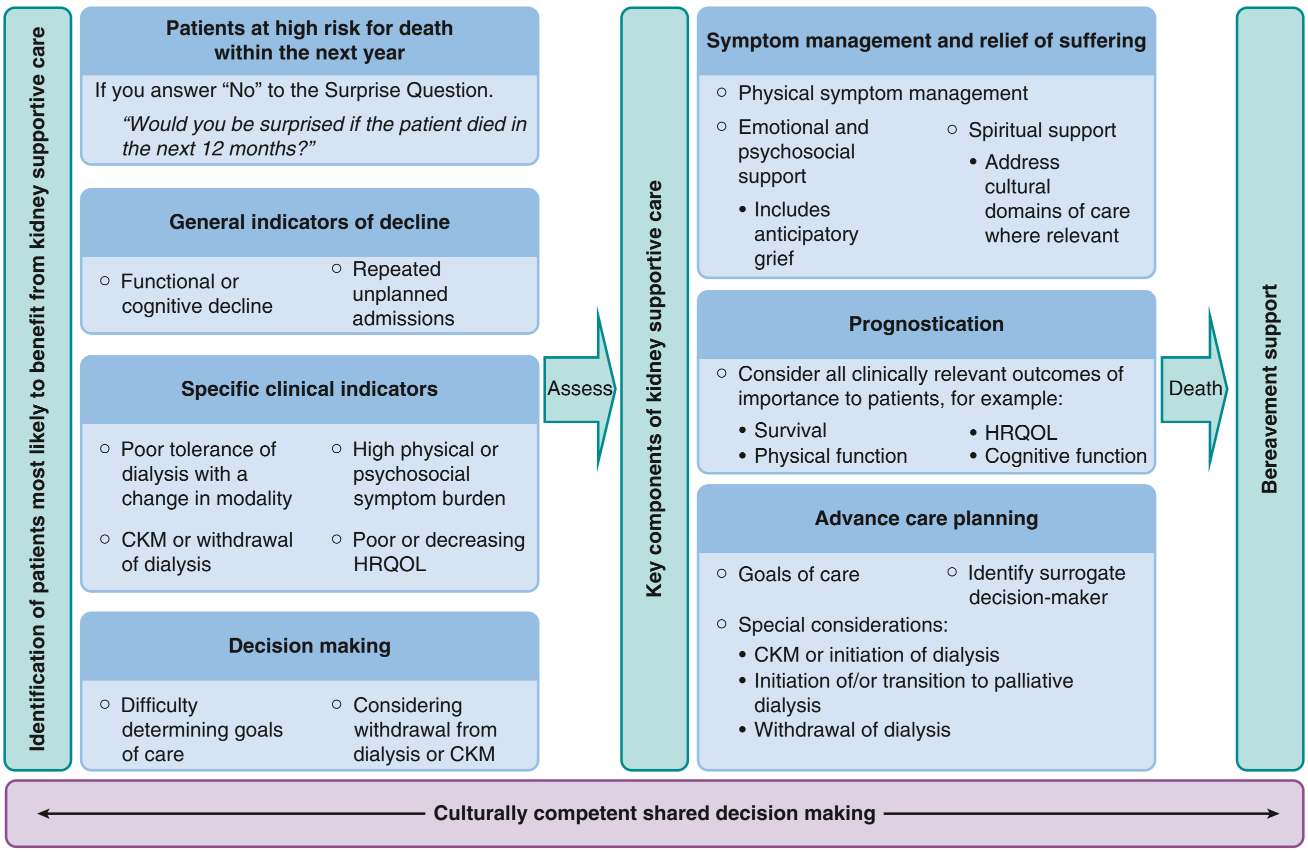

Fig. 61.2 An approach to kidney supportive care. *CKM,* Conservative kidney management; *HRQOL,* health-related quality of life. (Adapted from Davison SN. Integrating palliative care for patients with advanced chronic kidney disease: recent advances, remaining challenges. *J Palliat Care*. 2011;27:53–61.)

In many parts of the world, the cultural norm is protection of the patient from the truth. This often involves cultural beliefs surrounding the cause and meaning of illness. This can complicate decision making, especially in the context of sharing prognosis. For example, some Aboriginal and Asian cultures prohibit explicit references to dying based on an interpretative framework in which language has the capacity to create reality.[25] Positive thinking is felt to promote health, whereas the delivery of bad news could shorten the life of the patient. Family members may prefer to communicate prognostic information themselves in such a way as to "balance" hope with the bad news. Health care providers need to understand that in some contexts, this may be appropriate.

Cultural views can impact how patients want care to be delivered. Many cultures possess little knowledge of, or have little exposure to, palliative care and may be reluctant to receive this care without understanding the direct benefit to them. Cultural values may also determine who should and who should not be directly involved in providing physical care, with many cultures strongly preferring that family be directly involved.[26] Many non-Western cultures such as traditional Chinese and indigenous cultures view the body, soul, and spirit as an integrated whole in the context of strong interpersonal relationships and may require access to a traditional healer. This may be viewed as similar to access to a hospital chaplain in a Western context.[27]

Many people from rural and remote areas wish to die at home connected to land and family for strong cultural reasons yet are relocated for end-of-life care. For indigenous people, relocation at the end of life is an extremely frightening experience.[28] Culture also plays a role in the expression of suffering and grief, with many cultures being reluctant to complain of pain. Others are not comfortable with frank, direct styles of communication and therefore may not express suffering or grief overtly as this may be seen as inappropriate. In these cases, emotional containment does not mean indifference or a lack of suffering.

Culturally competent shared decision making requires that clinicians and patients jointly consider best clinical evidence in light of a patient's specific health characteristics and values within the framework of the patient's and family's cultural beliefs. In increasingly diverse societies, the challenge for health care professionals is to understand how these numerous cultural differences shape the care for an individual patient. Cross-cultural communication strategies required to address this have recently been explored[29,30] and are summarized in Table 61.1. Cross-cultural communication starts with understanding one's own beliefs, values, and experiences and then acknowledging individual cultural traditions, which may be sharply divergent. Health care professionals also need to recognize the diversity of beliefs and practices within cultural groups. The process of acculturating may complicate this, and many people will hold blended cultural perspectives. Health care professionals therefore must be careful not to assume the preferences of individuals on the basis of their cultural group. When uncertain about how a patient or family perceives a situation, it is best simply to ask.[29,30]

PROGNOSTICATION

Estimating and communicating prognosis is required for shared decision making.[1] Being able to prognosticate accurately helps health care professionals identify high-risk patients, facilitate discussions regarding the need for targeted KSC services, and align care pathways to best meet an individual's needs. Given that KSC focuses on the individual patient's priorities, patient-specific prognoses for relevant outcomes beyond survival, such as the impact of treatments on HRQOL, physical function, hospital-free survival, and symptoms, are needed.

Studies have shown that despite patients wanting to discuss their prognosis, it is often not done.[5,31] Prognostication is challenging in CKD given the great variation in individual illness trajectories.[32] Accurate prognostic models are limited, and nephrologists struggle to explain illness complexity and predict clinical trajectories.[33,34] Although there is a large volume of data on prognostic markers, individually these factors are often of little practical relevance. Only a small number of studies have attempted to combine these factors into clinically useful prognostic tools. A recent review highlights the limits and deficiencies of currently available prognostic tools,[35,36] which are outlined in Table 61.2. These tools are limited to progression of kidney failure and survival on dialysis. They do not prognosticate outcomes for people receiving CKM, nor do they address patient or family concerns of HRQOL, function, and symptom burden.

As the field of prognostication moves forward, new markers may enhance current approaches. These may include identifying a decline in physical function using tools, such as gait speed, a modified Karnofsky activity scale, activities of daily living, or frailty scores.[45-51] Repeat hospitalizations,[52] a decline in HRQOL scores,[53] reduced nutritional scores,[54] reduced appetite,[55] and lower body weight[56,57] may all be simple and reliable means for independently identifying CKD patients at risk for early death or poor health outcomes. Ideally, these prognostic tools would be used alongside patient decision aids to help in the shared decision-making process. Accurate prognostication will likely provide better opportunities to meet patients' and families' needs in the final years and months of life. New models, however, will need to be evaluated in different clinical and cultural contexts.

SYMPTOM ASSESSMENT AND MANAGEMENT

Symptom assessment and management are integral components of KSC. Advanced CKD is associated with a high symptom burden, and patients often experience complex clusters of symptoms, such as pruritus, pain, restless legs, fatigue, anorexia, nausea, insomnia, anxiety, and depression.[58-60] Research suggests that symptom burden is more important than objective clinical parameters in determining HRQOL in people with CKD.[61] Symptom burden was shown to account for 29% to 44.6% of the impairment in dialysis patients' physical HRQOL scores and 39% to 48.7% of their impairment in mental HRQOL scores.[58-60] Although dialysis treatment may address some

Table 61.1 Strategies for Culturally Competent Shared Decision Making

Task	Communication Strategy	Additional Considerations
Understand one's own beliefs, values, and experiences	Having an awareness of individual differences from that of the patient is important to create a respectful and nonbiased dialogue. It will also help resolve conflict when differences with patients' beliefs and wishes arise.	Consider one's perspective as a clinician and the influence of the health care system and institution within which one delivers care.
Understand the patient's experience	Ask about the patient as a person, where they are from, and degree of integration within the ethnic community. Assess how the patient interprets his or her condition (i.e., the cause and impact).	This includes asking what their most important concerns are now that they have this illness. This establishes the groundwork for negotiating mutually acceptable goals for care.
Acknowledge individual cultural traditions	Ask about the patient's health beliefs, values, practices, and cultural communication etiquette. This includes attitudes toward truth telling. Avoid generalizing a patient's beliefs or values on the basis of cultural norms.	The "RISK" reduction assessment is a helpful strategy to ascertain the level of cultural influence for a patient.[30]
Giving information	**Ask–Tell–Ask** (this strategy can extend to many of the tasks described here): **Ask** the patient's understanding of his or her illness, what kind of information the patient and/or family needs, and how they want the information told. **Tell** (give) information concisely, avoid jargon, and disclose only one to three pieces of information at any given time. **Ask** what the patient understands or will take away from the conversation.	The final ask ensures the information has been understood and invites the patient and family to share concerns or explore lingering questions. Are there language barriers or low health literacy barriers that may hinder understanding and shared decision making?
Determine the level of patient engagement in decision making	Assess preferences for decision making and who will be involved.	Explicitly explore preferences for family-centered versus individual-centered decision making.
Address trust concerns	Be nonjudgmental, transparent, and avoid defensiveness.	The goal is to create an atmosphere of mutual respect and avoid misunderstanding.
Address resources needs	Ask what tangible resources they need to navigate the health care system. Ask if assistance is available to them and their family in their community.	This may be influenced by language, level of education, socioeconomic status, and their social support networks.

From Brown EA, Bekker HL, Davison SN, et al. Supportive care: communication strategies to improve cultural competence in shared decision making. *Clin J Am Soc Nephrol*. 2016;11:1902–1908.

uremic symptoms, it may do little to address symptom burden and HRQOL in more frail patients and chronic inflammation, malnutrition, and frailty continue to progress regardless of whether dialysis is started or not. At a population level, 1 year of dialysis did not result in an improvement in overall symptom burden or HRQOL.[62] The patients who tend to do the best are those with limited comorbidity; they tend to have low symptom burden until shortly before needing dialysis and experience the more typical uremic symptoms of anorexia, nausea, vomiting, and fatigue, and these can quickly improve after starting dialysis. Kidney Disease: Improving Global Outcomes recommends incorporating regular global symptom screening using validated tools such as the ESAs-r:Renal (Edmonton Symptom Assessment System: Renal) into routine clinical practice and the incorporation of systematic, stepwise approaches to managing these symptoms.[1,63]

Many global symptom assessment tools of varying length have been used in CKD studies.[64] For clinical utility it is important that these tools are valid, reliable, and sufficiently short and simple to minimize patient and staff burden. They must be appropriate for use in high-risk patients who are frail with cognitive impairment. In the case of pain, 0- to 100-mm visual analog scales or 0 to 10 numeric rating scales are advised based on extensive literature around what constitutes clinically important differences in pain intensity.[65] Core domains that should be assessed when managing pain include pain intensity, function, HRQOL, and adverse events.[65,66]

Several self-reported HRQOL measures have also been used for CKD patients. Some are generic measures, whereas others are disease specific such as the Kidney Dialysis Quality of Life (KDQOL) Questionnaire and the shorter version (KDQOL-SF).[67] These multidimensional tools focus

Table 61.2 Prognostic Tools Relevant to Kidney Supportive Care

Data Source	Population Studied	Parameter	C-Statistic
Progression of CKD to Kidney Failure Defined as Need for Dialysis or Preemptive Kidney Transplantation: Kidney Failure Risk Equation			
Two Canadian cohorts[37,38] A total of 31 multinational cohorts from the CKD Prognosis Consortium	Development and initial validation with 3449 and 4942 Canadian patients with G3-5 CKD, respectively. Additional validation in 721,357 multinational participants with G3-5 CKD	Four-variable model: age, gender, eGFR, and ACR Currently available for free online (https://www.qxmd.com/calculate/calculator_308/kidney-failure-risk-equation-4-variable)	Pooled data from the 31 multinational cohorts 2-year prediction: 0.9 (95% CI 0.89-0.92) 5-year prediction: 0.88 (95% CI 0.86-0.90)
		Eight-variable model: age, gender, eGFR, ACR, calcium, phosphate, bicarbonate, and albumin	Pooled data from the 31 multinational cohorts 2-year prediction: 0.89 (95% CI 0.88-0.91) 5-year prediction: 0.87 (95% CI 0.85-0.88)
3 Months' Survival after Dialysis Start			
USRDS[39]	69,441 incident patients aged ≥67 years	Age, albumin, assistance with ADL, nursing home, cancer, heart failure, and hospitalization Currently available for free online (https://qxmd.com/calculate/3-month-mortality-in-incident-elderly-esrd-patients)	0.681 (95% CI 0.676-0.686)
		A more comprehensive version adds gender, race, central vein dialysis catheter use, early nephrology referral, albumin, creatinine, peripheral vascular disease, and alcohol abuse	0.712 (95% CI 0.706-0.718)
French REIN registry[40]	28,496 incident patients aged ≥75 years	Gender, age, congestive heart failure, severe peripheral vascular disease, dysrhythmia, severe behavioral disorders, active malignancy, serum albumin, and impaired mobility	0.749 (95% CI 0.743-0.755)
Catalan[41] renal registry	1365 incident diabetic adult patients	Age, functional autonomy, heart disease, and central catheter as vascular access	0.77 (95% CI 0.742-0.798)
6 Months' Survival after Dialysis Start			
French REIN[42] registry	4142 incident patients aged ≥75 years	BMI, diabetes, congestive heart failure, peripheral vascular disease, dysrhythmia, active malignancy, severe behavioral disorder, dependency for transfers, and initial context of dialysis start	0.70 (95% CI 0.671-0.729)
6 Months' Survival on Hemodialysis			
New England HD clinics[43]	1026 adult-prevalent patients. External validation in 1372 prevalent patients[44]	Age, dementia, peripheral vascular disease, serum albumin, and the "surprise question" Currently available for free online (https://qxmd.com/calculate/calculator_135/6-month-mortality-on-hd)	0.80 (95% CI 0.73-0.88)

ACR, Urine albumin-to-creatinine ratio; *ADL,* activities of daily living; *BMI,* body mass index; *CI,* confidence interval; *CKD,* chronic kidney disease; *C-statistic,* concordance (C)-statistic (which corresponds to the area under receiver operating characteristic curve, ranges from 0 to 1 depending on accuracy of discrimination); *eGFR,* estimated glomerular filtration rate; *G3-5,* estimated glomerular filtration rate category 3-5; *HD,* hemodialysis; *REIN,* Renal Epidemiology and Information Network; *USRDS,* United States Renal Data System.

on physical and emotional symptoms, burden of disease and effects on daily life, cognitive function, work status, sexual function, quality of social interaction and social support, staff encouragement, and patient satisfaction. Unfortunately, they are burdensome, requiring interviewer assistance and substantial time to complete. Although such tools provide comprehensive HRQOL information, they are perhaps more suited to a research environment where dedicated staff can help with the administration and complex scoring. To embed a HRQOL measure into routine clinical care, simple tools that can be interpreted easily by staff will most likely be required.[63,68,69] Generic and disease-specific measures have been shown to perform similarly in assessing changes in HRQOL in people on hemodialysis,[70] and an Oxford review of patient-reported outcome measures in CKD recommended EQ-5D-5L for use in this patient population.[71] Recommendations for symptom assessment tools that can be used in routine clinical care are outlined in Table 61.3.

The aim of treatment is to ameliorate symptoms that are burdensome and adversely impact the patient's HRQOL as it is not always necessary or possible to resolve them completely. It is important to acknowledge this and negotiate with the patient an acceptable level of symptom control. For pain, there is good evidence to suggest that targeting a pain of <33/100 mm on a visual analog scale or 3/10 on a numeric rating scale or a change in pain intensity of at least 30% (moderate benefit) or 50% (substantial benefit) will deliver worthwhile HRQOL benefits.[66] Given the synergistic and interrelated nature of symptoms experienced in advanced CKD, an approach to care that addresses overall symptom burden is likely to improve HRQOL even if each individual symptom has not resolved completely. For example, a moderate reduction in pain may be sufficient to improve sleep, improve mood, and increase the ability to cope with health challenges, resulting in a substantial improvement in function and HRQOL. Recommendations for the management of common symptoms in CKD are outlined in Table 61.4. These recommendations are appropriate for people with kidney failure who have yet to start dialysis, people who are receiving CKM, and for people on hemodialysis.[77,78] A general stepwise approach to symptom management involves 1. ruling out contributing factors, 2. maximizing the use of nonpharmacologic interventions, and 3. considering pharmacologic interventions if symptoms continue to impact adversely the patient's HRQOL.

CHRONIC PAIN MANAGEMENT

Although detailed information on the management of chronic or persistent pain is beyond the scope of this chapter, some general principles are worth highlighting. Chronic pain can be defined as any painful condition that persists for more than 3 months. Dialysis patients may also experience recurring episodes of acute pain, such as pain from needling fistulas or intradialytic steal syndrome, intradialytic headaches, and cramps. These acute pains tend to be associated with tissue damage but typically have no progressive pattern, last a predictable period, subside as healing occurs, and are episodic with periods without pain. By contrast, chronic pain is present for long periods and is often out of proportion with the extent of the originating injury. It is more likely to result in functional impairment and disability, psychological distress, sleep deprivation, and poor HRQOL.

Nonpharmacologic therapies are a vital part of managing chronic pain and are typically required to augment pharmacologic treatments to achieve adequate relief (i.e., multimodal therapy). They may also be used as stand-alone therapies. Examples are outlined briefly in Table 61.4. Core principles in developing a treatment plan for chronic pain include explaining the nature of the chronic pain condition, setting appropriate goals, and developing a comprehensive treatment approach and plan for adherence. Medication should not be the sole focus of treatment and should only be used when needed, in conjunction with other nonpharmacologic modalities, to meet treatment goals. Optimal patient outcomes often require multiple approaches used in concert, coordinated via a multidisciplinary team.

There are five essential principles for the pharmacologic management of pain. These are described in Table 61.5. Of particular importance is the careful selection of analgesics when prescribing for people with advanced CKD. Many analgesics and their metabolites are excreted by the kidney through glomerular filtration, tubular secretion, or both. People with CKD are at risk for accumulation of toxic metabolites if not monitored carefully. The choice of an appropriate initial therapeutic strategy depends on an accurate evaluation of the cause of the pain and the type of chronic pain syndrome. In particular, neuropathic pain should be distinguished from nociceptive pain (Table 61.6). For most patients with advanced CKD, the initial treatment of neuropathic pain involves calcium channel $\alpha2$–δ ligands (e.g., gabapentin and pregabalin) or tricyclic antidepressants (see Table 61.4). Opioid medications are considered second-line agents for neuropathic pain. By contrast, the pharmacologic approach to nociceptive pain primarily involves nonopioid and opioid analgesics if nonpharmacologic strategies are insufficient. The opioids that are generally considered safer for use in people with advanced CKD are hydromorphone, buprenorphine, fentanyl, and methadone, with doses started low, titrated slowly, and the patient monitored carefully for analgesia, adverse effects, overall function, and HRQOL (see Table 61.4).

SPECIAL CONSIDERATIONS AT THE END OF LIFE

As a patient's condition deteriorates, certain nonpharmacologic interventions will become less realistic (e.g., exercise). Energy conservation and restoration will become of utmost importance. Ensuring that appropriate support is in place to assist with activities of daily living and that nursing care is available as needed is essential. Drowsiness may increase as the end of life approaches due to disease progression and/or medications. Some patients and families may even prefer increased sleepiness if this helps ensure comfort. Symptoms should be anticipated with the appropriate prescriptions in place to address a patient's symptoms as they appear, including an alternative route to oral administration as swallowing becomes compromised.

Table 61.3 Recommendations for Clinical Symptom Assessment Tools for Use in People With Advanced Chronic Kidney Disease

Description	Clinical Utility
Global Symptom Assessment	
Edmonton Symptom Assessment System–Revised: Renal (ESAS-r: Renal)[58,59,72]	
A 0-10 NRS for 12 symptoms: pain, activity, nausea, depression, anxiety, drowsiness, appetite, well-being, shortness of breath, pruritus, sleep, and restless legs. It also has a place to capture "other problems." The scale for each symptom is anchored by the words "no" and "severe" at 0 and 10, respectively. The sum of all scores makes up the overall symptom distress score ranging from 0-120.	This is a short, practical tool for global symptom screening, which can be rapidly and repeatedly completed by patients and therefore incorporated easily into routine clinical care, even for patients who are preterminal. It has been translated into several languages. It uses the 0-10 NRS advised for the assessment of pain intensity.
Palliative Care Outcome Scale–Renal (POS-Renal)[73]	
Assesses 17 symptoms: pain, shortness of breath, weakness or lack of energy, nausea, vomiting, poor appetite, constipation, mouth problems, drowsiness, poor mobility, itching, difficulty sleeping, restless legs or difficulty keeping legs still, anxiety, depression, changes in skin, and diarrhea. These are rated in terms of their impact on the patient over the past week using a 0 (not at all) to 4 (overwhelmingly) Likert scale.	This tool is slightly longer than the ESAS-r: Renal but is simple to use. It has been translated into several languages. A disadvantage is that is does not use the 0-10 NRS that is advised for assessing pain and does not assess fatigue, which is one of the most prevalent and bothersome symptoms for people with advanced CKD. The concept of weakness is not the same as fatigue.
Multidimensional Pain Assessment	
The Brief Pain Inventory (BPI)[74]	
Assesses the location, type (nociceptive vs. neuropathic), and intensity of pain. It also evaluates the impact of pain on the core domains of function and HRQOL. Specifically, it explores the impact of pain on general activity, mood, walking ability, work, relationships, sleep, and enjoyment of life. The standard 32-question instrument has been condensed to a 9-question short form.	This tool has been used successfully in clinical and research settings internationally to assess pain once identified as a problem. The short form is simple to use with minimal respondent burden, and seriously ill patients have been successful in completing it. The Interference Scale of the BPI has been recommended by IMMPACT for the assessment of physical functioning, one of the core pain assessment domains.[75] A change of 1 point on the Interference Scale is considered the minimally clinically important change.[65,66] It has been translated into several languages.
Health-Related Quality of Life Assessment	
5-Level EuroQol 5 Dimension (EQ-5D-5L)[76]	
This generic HRQOL measure has 2 parts: 1. EQ-5D descriptive system and 2. EQ VAS. The descriptive system comprises 5 dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Each dimension has 5 levels: "no problems," "slight problems," "moderate problems," "severe problems," and "extreme problems." The EQ VAS records the patient's self-rated health on a vertical 0- to 100-mm VAS, where the endpoints are labeled "The best health you can imagine" and "The worst health you can imagine."	This short, practical tool for assessing HRQOL is available in more than 130 languages.

CKD, Chronic kidney disease; *HRQOL,* health-related quality of life; *IMMPACT,* Initiative on Methods, Measurement, and Pain Assessment in Clinical Trials; *NRS,* numerical rating scale; *VAS,* visual analog scale.

Fig. 61.3 outlines a terminal care algorithm for pain, which can also be used for breathlessness.[77] As a general rule, benzodiazepines should be avoided in patients with advanced CKD, especially those who are functionally impaired and/or more frail. Benzodiazepines carry significant risks including an increased risk of falls, fractures, and decreased cognition. However, they may be the only option for a patient who can no longer swallow and may be beneficial for restless legs, anxiety, and agitation near the end of life. Fig. 61.4 describes a terminal care approach to restlessness, agitation, and delirium.[77]

ADVANCE CARE PLANNING

The unpredictable nature of CKD progression and associated complications make it particularly important for patients and families to plan for end-of-life care needs.

Table 61.4 Symptom Management for People with Advanced Chronic Kidney Disease[77,78]

Restless Legs Syndrome

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
Anemia Iron deficiency Hyperphosphatemia Medications such as dopamine antagonists, antidepressants, and opioids (some of these drugs are commonly prescribed at end of life, e.g., haloperidol and opioids)	Abstinence from stimulants (e.g., alcohol, caffeine, and nicotine) Mental alerting activities (e.g., puzzles and games) Good sleep hygiene Exercise	First line: gabapentin (50-300 mg daily) 2-3 hours before sleep, especially if concomitant pruritus, insomnia, and/or neuropathic pain are reported Second line: nonergot-derived dopamine agonists (pramipexole 0.125 mg daily, ropinirole 0.25 mg daily, rotigotine transdermal patch 1-3 mg) given 2 hours before sleep At the end of life if swallowing is problematic: consider midazolam 1 mg subcutaneously q4h PRN	The most common side effects of gabapentin are drowsiness, dizziness, confusion, fatigue, and occasionally peripheral edema Nonergot-derived dopamine agonists have shown success in reducing symptoms in idiopathic RLS, but there are limited data in uremic RLS. Side effects might include headache, insomnia, and nausea. Augmentation may occur with long-time use. Benzodiazepines are not a first-line treatment for RLS, but there is some limited evidence for their use. If the patient is experiencing refractory RLS causing significant sleep disturbance or if benzodiazepines may potentially treat concurrent symptoms (e.g., anxiety), they could be considered.

Uremic Pruritus

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
Anemia Iron deficiency Hyperphosphatemia Hypercalcemia Other: xerosis, drug hypersensitivities, allergies, infestations, contact dermatitis, or inflammation	Good skin care and moisturizers (e.g., baths with lukewarm water, pat dry and moisturize within 2 minutes; gentle soaps with no fragrances or additives) Keep skin cool Humid environment Avoid scratching—keep fingernails short, encourage gentle massage, and wear gloves at night Consider complimentary therapies (e.g., phototherapy [UVB] 3 times weekly for a 3-week trial; acupuncture. Little evidence exists for these alternative therapies.	*Topical* Capsaicin 0.025% or 0.03% ointment Pramoxine 1% Menthol/camphor/phenol—0.3% each γ-Linolenic acid cream 2.2%	*Topical* These agents can be applied two times daily (four times daily for capsaicin). Capsaicin may cause burning to the area initially. Menthol, camphor, and phenol are separate products that can be added to most creams. All 3 may be added together, commonly with a 0.3% concentration for each.
		Systemic First line: gabapentin (50-300 mg daily) 2-3 hours before sleep. Second line: tricyclic antidepressant such as doxepin 10 mg daily at night	*Systemic* The most common side effects of gabapentin are drowsiness, dizziness, confusion, fatigue, and occasionally peripheral edema. Potential adverse effects of tricyclic antidepressants include dizziness, blurred vision, constipation, and urinary retention. There is an increased risk of confusion and sedation, particularly in older adults.

Continued

Table 61.4 Symptom Management for People with Advanced Chronic Kidney Disease[77,78]—cont'd

Nausea and Vomiting

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
Metabolic disturbances (e.g., uremia) Medications (e.g., opioids and SSRI antidepressants) Gastrointestinal disturbances (e.g., constipation and delayed gastric emptying)	Manage constipation Encourage good oral hygiene Smaller, more frequent meals; eat meals slowly Avoid alcohol Avoid foods that are greasy, spicy, or excessively sweet Minimize aromas (e.g., cooking odors, perfumes, and smoke) Encourage relaxed, upright position after eating to facilitate digestion Loose-fitting clothing Consider complementary therapies (e.g., relaxation techniques, acupressure, and use of ginger)	First line: Ondansetron 4-8 mg every 8 hours as needed Second line: metoclopramide 2.5 mg every 4 hours as needed (first-line therapy if nausea is due to delayed gastric emptying) Third line: olanzapine 2.5 mg every 8 hours as needed OR haloperidol 0.5 mg every 8 hours as needed Fourth line: for persistent and severe nausea, consider increasing haloperidol to 1.0 mg (maximum 5 mg in 24 hours) OR replacing with methotrimeprazine 5 mg orally or 6.25 mg subcutaneously every 8 hours as needed	Ondansetron can be constipating. Haloperidol, metoclopramide, and olanzapine are all dopamine antagonists: Avoid prescribing them together. They can also exacerbate RLS. They all cross the blood-brain barrier, and extrapyramidal symptoms are possible. Haloperidol has a higher risk of extrapyramidal symptoms than metoclopramide and olanzapine. Increasing the dose of methotrimeprazine may lead to levels of drowsiness that the patient may find unacceptable and should be discussed with the patient and/or family

Breathlessness

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
Anxiety Anemia Infection Volume overload leading to pulmonary edema	Sit in an upright position (e.g., 45 degrees) Position by a window or use a fan to blow air gently across the face Maintain a humid environment Pursed lip breathing Supplemental oxygen Complementary therapies (e.g., relaxation techniques, and music) Consider role of diet such as sodium and fluid restriction if patient is volume overloaded	If patient is intravascularly volume overloaded: diuretic such as a loop diuretic—furosemide. Occasionally patients may require combination diuretic therapy—consider adding metolazone Near the end of life: low doses of opioids are the most effective treatment **For breathlessness that is episodic** and primarily associated with a specific activity, consider fentanyl 12.5 μg subcutaneously or sublingually PRN **For shortness of breath that is more constant or unpredictable** in nature, consider hydromorphone 0.5 mg PO (0.2 mg subcutaneously) every 4 hours around the clock and every hour as needed	Due to accumulation of metabolites, opioids should always be started at a low dose and monitored closely for adverse effects Due to its fast action, fentanyl works well in cases where breathlessness is predictable

Table 61.4 Symptom Management for People with Advanced Chronic Kidney Disease[77,78]—cont'd

Fatigue and Sleep Disturbances

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
Fatigue Vitamin D deficiency Metabolic acidosis Hyperphosphatemia Secondary hypothyroidism Anemia Malnutrition Mood disorders Sleep disturbances Sleep disturbances Other symptoms (e.g., restless legs, pruritus, pain, and breathlessness) Cognitive impairment Medications Generalized insomnia Mood disorders Sleep apnea	Fatigue Exercise Nutrition and hydration management Energy-conservation strategies Good sleep hygiene (e.g., avoid stimulants before bed, avoid napping during the day, and save the bedroom for sleep) Cognitive and psychological approaches (e.g., relaxation therapy, delegating, and setting limits) Complementary treatments (e.g., acupressure, massage, and acupuncture)	Sleep disturbances First line: consider low-dose gabapentin (50-300 mg at night), especially if the patient has concomitant neuropathic pain, RLS, or uremic pruritus Second line: Doxepin 10 mg at bedtime, especially if concomitant pruritus or neuropathic pain reported Third line: cautiously consider mirtazapine 7.5 mg or zopiclone 3.75-5 mg at night or melatonin 2-5 mg at night	Reassess medications after 2-4 weeks. Avoid over-the-counter sleep aids and benzodiazepines if possible. Specifically, avoid mirtazapine if taking tramadol or antidepressants. Monitor doxepin for anticholinergic side effects (e.g., dizziness, blurred vision, constipation, urinary retention, and cardiac arrhythmias). Evidence for melatonin is limited and inconclusive. Ideally, all of these medications should be prescribed for short-term use only

Nociceptive Pain

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
Determine cause for pain and consider appropriate investigations	Physical therapies (e.g., physical therapy, aerobic exercise, stretching, massage, acupressure, and acupuncture) Behavioral therapies (e.g., cognitive behavioral therapy—most commonly used behavioral therapy), biofeedback, relaxation techniques, psychotherapy/individual or group counseling, guided imagery, mindfulness-based stress reduction Interventional and surgical (e.g., ablative techniques, nerve blocks, and trigger point injections)	Step 1: Acetaminophen/paracetamol, maximum of 3 g daily. If pain is localized to a small joint, consider a topical NSAID (e.g., diclofenac gel 5% or 10% two to three times daily). Step 2: ADD an opioid to step 1 in very low doses. Hydromorphone starting at 0.5 mg PO (0.2 mg subcutaneously) every 4-6 hours; buprenorphine/fentanyl/methadone	Trial each step for 1-4 weeks before progressing, depending on pain severity Before starting an opioid, consider completing an opioid risk tool and order a bowel routine to avoid constipation (e.g., PEG 3350). All opioids should be started at low doses, monitored carefully for adverse effects and overall benefit, and titrated slowly (see Table 61.5)

Neuropathic Pain

Address Possible Contributing Factors	Nonpharmacologic Management	Pharmacologic Management	Additional Considerations
Determine cause for pain and consider appropriate investigations	As for nociceptive pain	Start with adjuvant therapy. First line: gabapentin, pregabalin (calcium channel $\alpha2$–Δ ligands) Second line: tricyclic antidepressants, amitriptyline starting at 10–25 mg daily or doxepin starting at 10 mg daily If more analgesia is required **in addition** to adjuvant therapy, add a nonopioid and then proceed with low dose of an opioid and titrate as described for nociceptive pain	Methadone may be effective for severe neuropathic pain because of its activity against the NMDA receptor antagonism

NMDA, *N*-methyl-D-aspartate; *NSAID*, nonsteroidal antiinflammatory drug; *PO*, per os; *q4h*, every 4 hours; *PRN*, as needed; *RLS*, restless legs syndrome; *SSRI*, selective serotonin reuptake inhibitor; *UVB*, short-wave ultraviolet B.

Table 61.5 Five Principles of Pain Management in Advanced Chronic Kidney Disease (CKD)

Principle	Description	Specific Considerations in Advanced CKD
"By mouth"	Oral administration is the safest and therefore preferred. Patient comfort and effectiveness must be considered. If ingestion or absorption is uncertain, analgesics need to be given by alternative routes, such as transdermal, rectal, or subcutaneous.	Hemodialysis patients have easy intravenous access. However, this is to be avoided as the route of administration for analgesics to optimize safety and minimize the risk of abuse and addiction.
"By the clock"	For continuous or predictable pain, analgesics should be given regularly. Additional "breakthrough" or "rescue" medication should be available on an "as needed" (PRN) basis in addition to the regular dose.	Some patients with mild pain may achieve adequate pain relief with analgesic dosing post-hemodialysis only. An example would be mild neuropathic pain dosed with gabapentin post hemodialysis.
"By the ladder"	Pharmacologic management proceeds stepwise from a nonopioid (step 1) to a very low dose of an opioid (step 2) using a modified WHO analgesic ladder. There are no good options for "weak" opioids for people with advanced CKD.	Careful selection of analgesics for each step of the ladder, taking into account the degree of kidney failure, is critical (see Table 61.4). Sustained-release preparations are generally not recommended in people with advanced CKD.
"For the individual"	There is large variability in patients' response to analgesics. The "correct" dose is the amount needed to relieve the pain without producing intolerable side effects. Evaluation and recording of benefits and toxicity are essential.	Chronic pain is often experienced in the context of numerous other physical, psychosocial, and spiritual concerns including end-of-life issues. Close attention to these other issues must not be forgotten as part of the pain management strategy.
"Attention to detail"	Pain changes over time; therefore there is the need for ongoing reassessment. Side effects should be explained and managed actively (e.g., constipation).	There are no studies on the long-term use of analgesics in people with CKD. Careful attention must be paid to efficacy and safety. The impact on overall symptom burden, physical function, emotional state, cognition, and HRQOL should be assessed routinely.

Table 61.6 Categories of Pain

	Neuropathic Pain	Nociceptive Pain
Description	Pain that results from damage to the nervous system resulting in either dysfunction or pathologic change	Pain that results from tissue damage in the skin, muscle, and other tissues, causing stimulation of sensory receptors
Pain characteristics	Common descriptors include burning, tingling, and shooting May be associated with episodes of spontaneous pain, hyperalgesia, and allodynia. The presence of the latter is pathognomonic. The pain may be felt at a site distant from its cause, in the distribution of a nerve, for example	Pain may be described as sharp, like a knife—often felt at the site of damage Pain may also be experienced as dull, aching, and poorly localized (as with stimulation of visceral nociceptors)
Response to analgesics	Generally responds poorly to analgesics and requires the use of adjuvant therapy	Generally responds well to analgesics

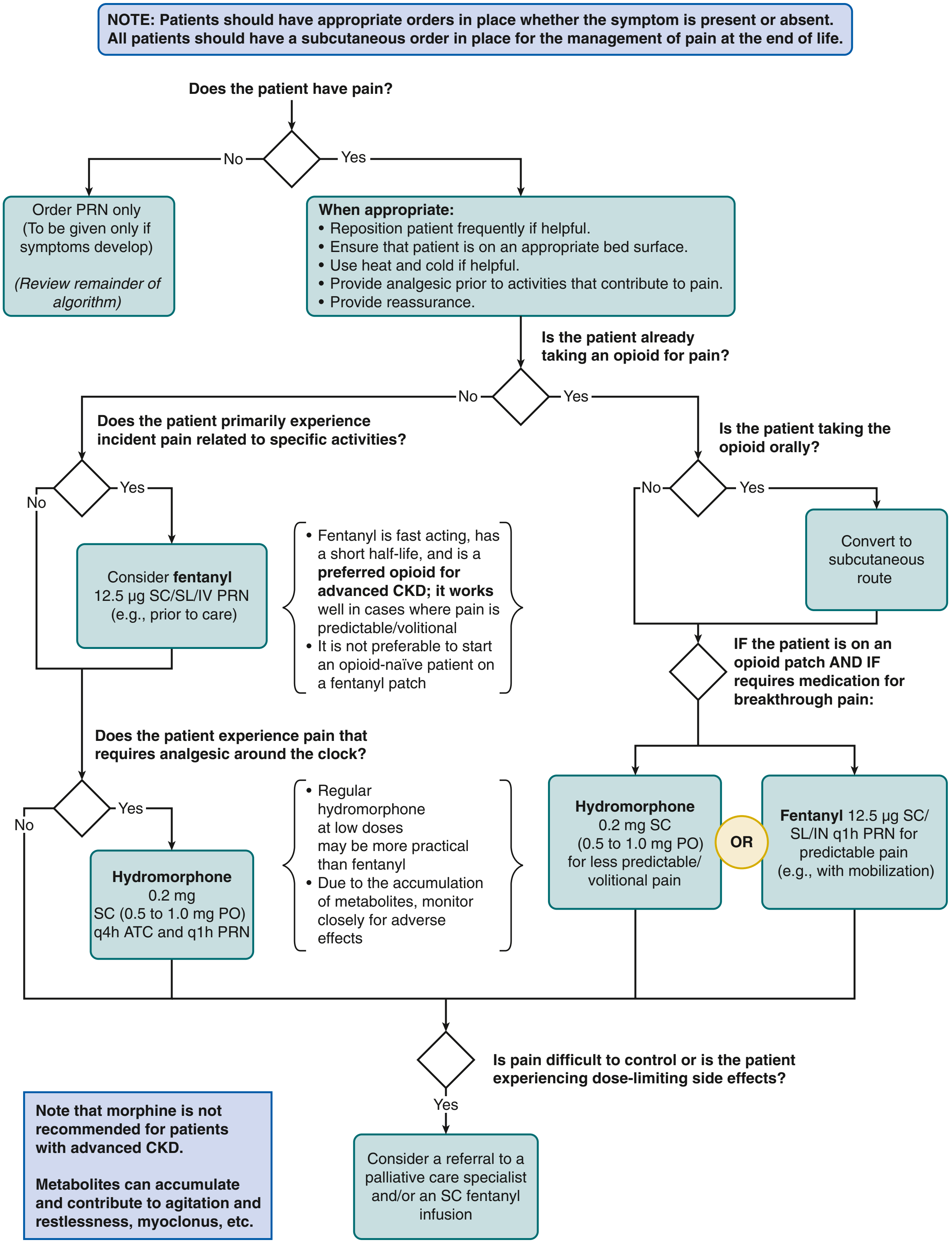

Fig. 61.3 Algorithm for the management of pain at the end of life for people with advanced chronic kidney disease. *ATC,* Around the clock; *PO,* per os; *q1h,* every 1 hour; *q4h,* every 4 hours; *SC,* subcutaneous; *SL,* sublingual. (Adapted with permission from Davison SN, Kidney Supportive Care Research Group. Conservative kidney management care pathway. <https://www.ckmcare.com>; 2016. Accessed 26.06.23.)

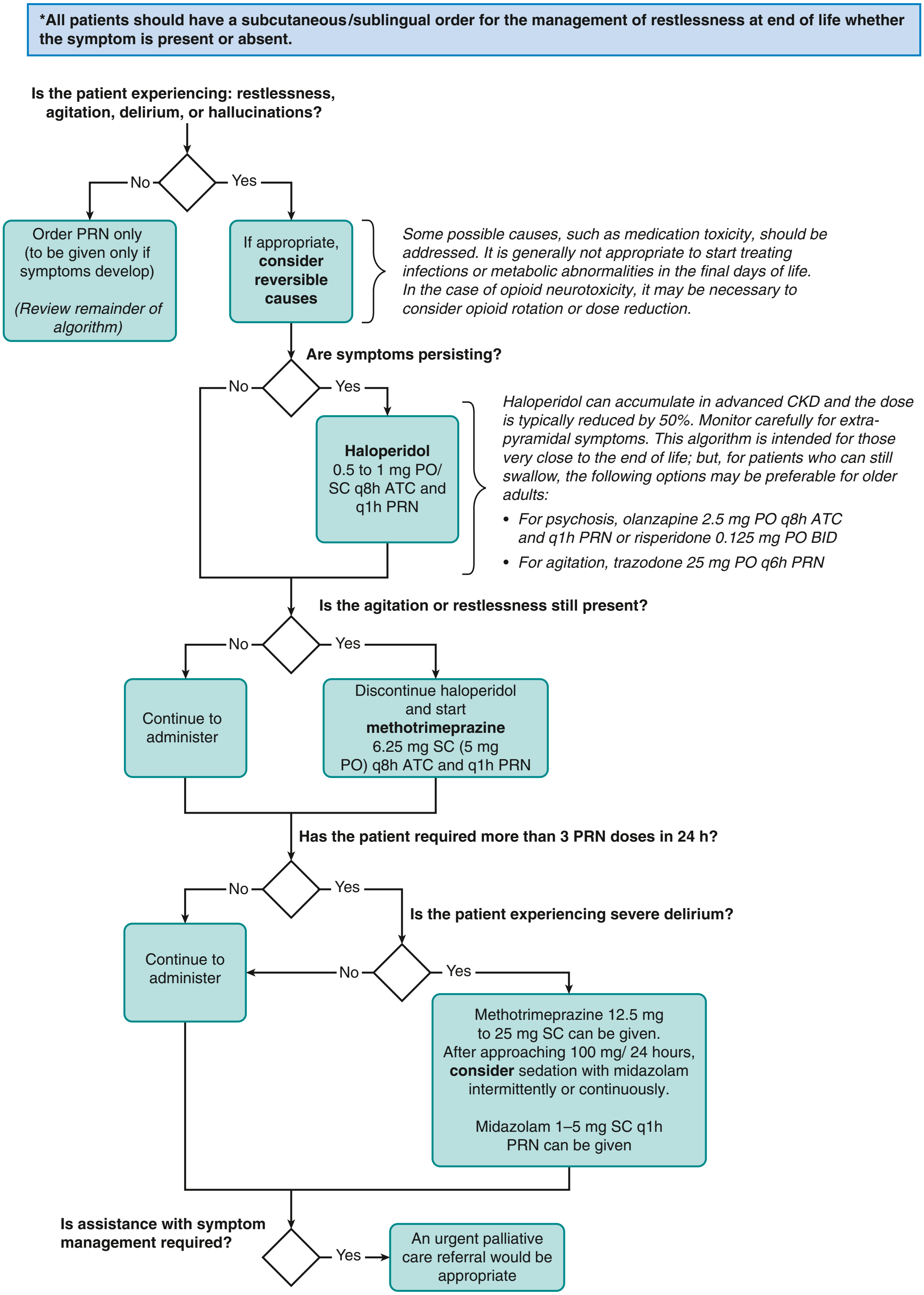

Fig. 61.4 Algorithm for the management of restlessness, agitation, and delirium at the end of life for people with advanced chronic kidney disease (CKD). *ATC,* Around the clock; *BID,* twice a day; *PO,* per os; *PRN,* as needed; *q1h,* every hour; *q6h,* every 6 hours; *q8h,* every 8 hours; *SC,* subcutaneous. (Adapted with permission from Davison SN, Kidney Supportive Care Research Group. Conservative kidney management care pathway. <https://www.ckmcare.com>; 2016. Accessed 26.06.23.)

Advanced care planning (ACP) is a process that involves understanding, communication, and discussion among a patient, the family (or other caregiver), and health care providers for the purpose of clarifying preferences for end-of-life care. It lays out a set of relationships, values, and processes for approaching end-of-life decisions for individual people when they can no longer speak for themselves including attention to ethical, psychosocial, and spiritual issues relating to starting, continuing, withholding, and stopping treatment such as dialysis.[1,79] The unique circumstances around treatment options for people with advanced CKD, such as palliative dialysis, CKM, and dialysis withdrawal, are discussed in greater detail later and further perspectives are also discussed in Chapters 60 and 80. When goals for care have not been established, patients often experience unnecessary hospitalizations, unwanted invasive treatments, a prolonged dying process, and ultimately the type of death they would not have wanted.[2]

An approach to ACP for patients with advanced CKD was developed on the basis of the perspectives of patients with advanced CKD[79,80] and is illustrated in Fig. 61.5. ACP is a health behavior, and these conversations need to be tailored to the readiness of the individual.[80] Behavior change theories such as the transtheoretical model or the "stages of change" model provide a useful framework for assessing patient readiness to engage in ACP.[81] This framework takes into account an individual's weighing of the pros and cons of engaging in ACP and assesses their attitudes about barriers to and facilitators of behavior change. A patient who is cognitively impaired or who is depressed or extremely anxious may have limited ability to participate meaningfully in ACP. The evidence suggests that a patient is much more likely to engage actively in ACP when he or she understands the personal relevance and perceived benefit.[80] Cultural, spiritual, and religious beliefs and variances around the concepts of autonomy, decision-making authority, communicating prognosis, and views around care at the end of life may influence attitudes toward ACP.[82] Clinicians should address any misconceptions the patient may have toward ACP and highlight the value of these conversations as it relates to that specific individual.

Several key elements to ACP conversations involve exploring the patient's understanding of their illness, past health care experiences, and values and beliefs (see Fig. 61.5). It is only through this exploration and understanding that the clinician will be able to align treatment options and end-of-life care with the patient's personal goals for living and dying well. It is not necessary that all elements be addressed in a single conversation but rather that over time both the patient and clinician gain a full understanding of the patient's beliefs, values, and overall goals for care. In routine clinical practice, it is common for these conversations to build on each other in subsequent visits. Although many clinicians feel they lack the skills and confidence to have ACP conversations with their patients,[31] effective ACP communication skills can be learned,[83-85] especially when using the strategies for culturally sensitive shared decision making outlined in Table 61.1.

Ideally, ACP should be integrated into routine kidney care early and addressed throughout the patient's illness. This helps to normalize the process; to afford patients the time to think, reflect, and make well-informed choices for future care; and to ensure care plans remain aligned with their preferences and prognosis as their health state declines. This often proves challenging as patients may feel that ACP is not yet relevant for them early in their disease. However, sudden illness or complications can occur without warning. At a minimum, however, high-risk patients should be identified through regular supportive care assessments as outlined in Fig. 61.2, to facilitate timely conversations with those most in need.

CONSERVATIVE KIDNEY MANAGEMENT

There is growing recognition that dialysis is unlikely to benefit and may even harm some patients, particularly those with significant preexisting functional or cognitive impairment and high levels of comorbidity.[86-88] Frail patients often experience accelerated functional and cognitive decline after starting dialysis.[89,90] Patients residing in long-term care facilities at the time of starting dialysis have especially poor outcomes.[89] CKM is an appropriate alternative to chronic dialysis for patients unlikely to benefit from dialysis (see also Chapter 60). CKM is defined as "care for people with kidney failure that focuses predominantly on providing KSC to promote HRQOL but does not include kidney replacement therapy."[12] CKM also includes interventions to delay progression and minimize complications of CKD in so far as doing so aligns with the individual's priorities.[12] Elderly patients with high comorbidity (especially ischemic heart disease), deceased functional ability, or age older than 80 years who receive CKM tend to live as long as patients who choose to start dialysis but with better preservation of function and HRQOL and fewer admissions to acute care settings, all while avoiding a burdensome treatment.[86-88,91] The international nephrology community recognizes that for patients unlikely to benefit from dialysis, dialysis should not be the default therapy and advocates strongly for the provision of quality CKM.[1,12]

Choosing CKM does not mean imminent death. Many patients live for months or years with kidney failure. Nor does CKM imply a "no-care" philosophy. CKM requires highly integrated communication and active care that involves tailor-made combinations of structures, processes, and interventions to address unique patient needs and unique system-community circumstances. Careful planning is necessary to link and transition patients at various points along the continuum of care, such as among primary, secondary, and tertiary care; between community-based and institutional care; and between acute care and long-term care. It may require involvement of mental health care and social services and should assist with referral to specialist palliative care and hospice services as available and appropriate. Multidisciplinary care teams will be required to deliver CKM. How best to implement CKM throughout complex, multisector health care systems is unclear. One method of implementation is through a clinical pathway that operationalizes best-practice guidelines into "how-to" steps for care delivery. A successful example of this is the interactive, online Canadian CKM pathway that includes an integrated plan of communication and care and a CKM patient decision aid that has been implemented across the province of Alberta. It has been shown to be effective in both

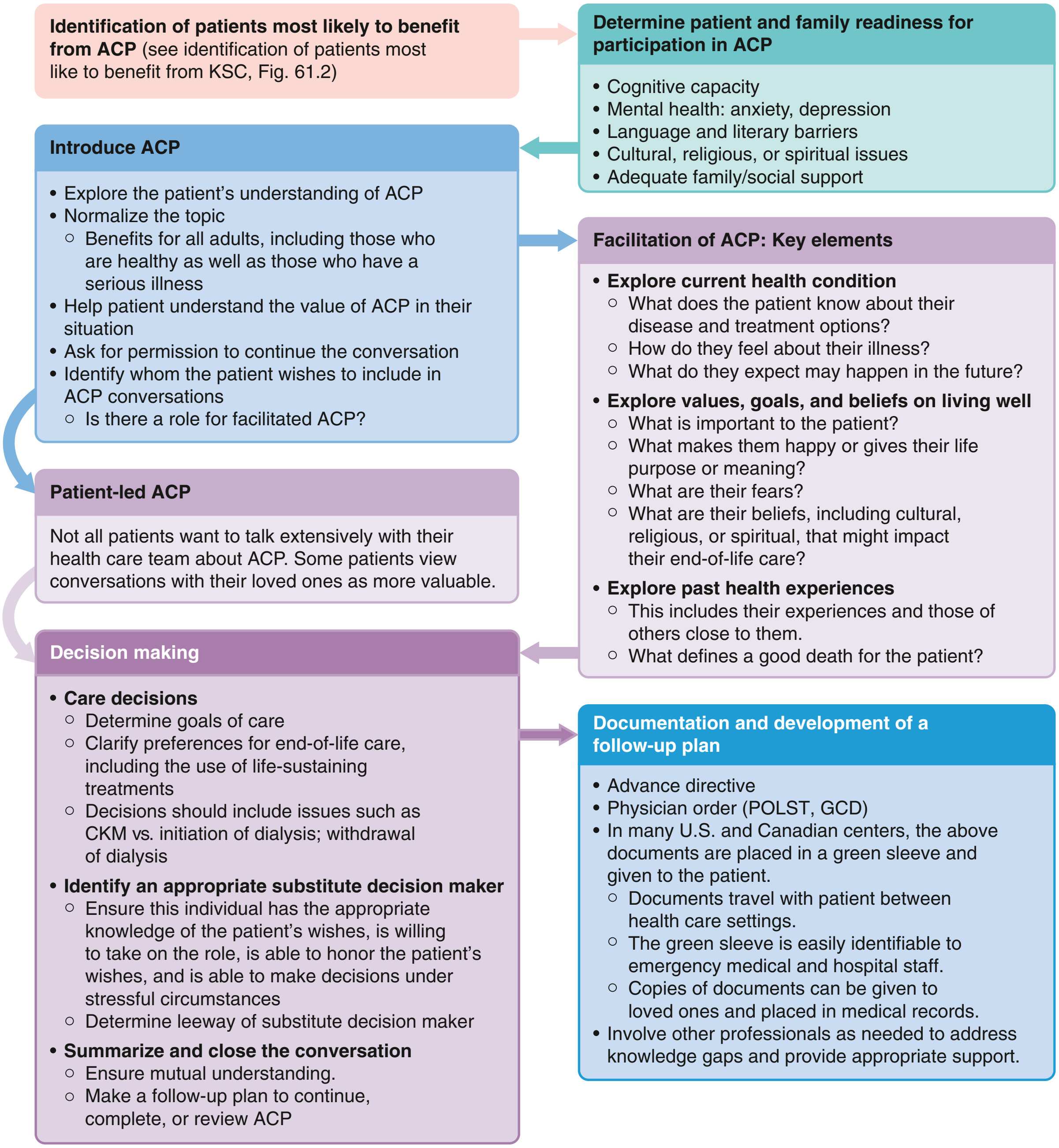

Fig. 61.5 Framework for facilitating advanced care planning *(ACP)* in patients with advanced chronic kidney disease. *CKM,* Conservative kidney management; *EOL,* end-of-life; *GCD,* goals of care designation; *KSC,* kidney supportive care; *POLST,* patient orders for life-sustaining treatment. (Adapted from Davison SN. Facilitating advance care planning for patients with end-stage renal disease: the patient perspective. *Clin J Am Soc Nephrol*. 2006;1:1023–1028; Davison SN, Torgunrud C. The creation of an advance care planning process for patients with ESRD. *Am J Kidney Dis*. 2007;49:27–36.)

increasing uptake of evidence-based care and optimizing patient management and outcomes (Fig. 61.6).[77,92]

Until recently, there were no clear recommendations for managing the metabolic complications of CKD for people receiving CKM. Current CKD guidelines were established for optimizing long-term health outcomes and do not address the priorities of care for patients who have chosen CKM or for those receiving dialysis who still want to prioritize HRQOL. CKM-specific recommendations for both CKD management and symptom control (as outlined in Table 61.4) have recently been suggested.[78] These recommendations and their associated rationales are outlined in Table 61.7. The illness trajectory of patients on a CKM pathway can be highly variable and is anticipated to pass through several stages. Patients with an estimated glomerular filtration rate (eGFR) of 10 to 15 mL/min/1.73 m^2 may remain relatively functional and stable for many months to years, whereas a patient who spends much of their time lying down with an eGFR of 5 mL/min/1.73 m^2 likely has a life expectancy measured in weeks to a few months. CKM guidelines need to consider the patient's general condition, prognosis, and goals of care. Earlier in the disease trajectory, maximizing

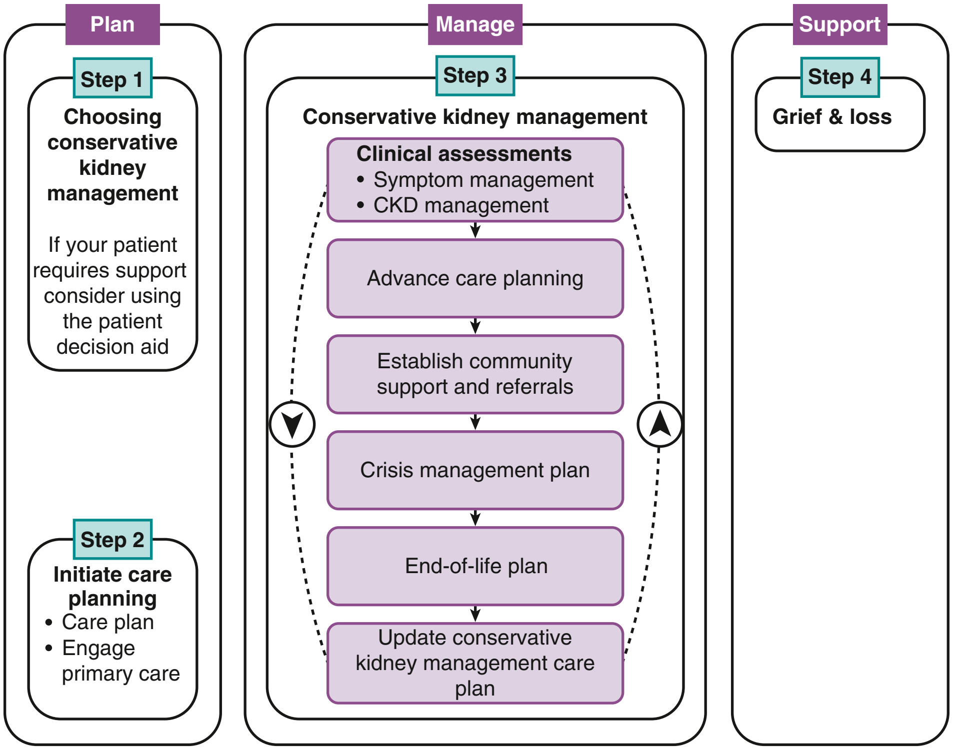

Fig. 61.6 An integrated approach to the care of people receiving conservative kidney management.[77,92] *CKD,* chronic kidney disease. (Reproduced with permission from Davison SN, Kidney Supportive Care Research Group. Conservative kidney management care pathway. <https://www.ckmcare.com>; 2016. Accessed 26.06.23.)

HRQOL likely requires a careful balance between promoting function and addressing symptom burden, whereas comfort and control of symptoms generally take precedence in the last weeks of life. There is minimal evidence specific to CKM; these recommendations have therefore incorporated both geriatric and palliative care principles and serve as a starting point for standardization of care that will need to be refined as evidence for best practice accumulates. By explaining the rationale and possible benefits and drawbacks of interventions, patients can become active participants in shared decision making, ensuring that all interventions remain aligned appropriately with their individual values, preferences, and prognosis.

CKM is nonexistent, poorly developed, and/or poorly integrated with kidney care in many settings across low-, middle-, and high-income countries[93] and decisions to start dialysis or CKM may be more reflective of physician preferences and/or lack of resources.[93,94] Recent studies report that in high-income countries where dialysis is readily available, most elderly patients feel they have no choice in the decision to start; are not informed fully of their options or the risks of dialysis; experience distress over unexpected complications and burdens of dialysis; desire more conversation about the decision; and rarely recall being given the option for CKM.[95,96] Many patients feel a sense of resignation to dialysis, especially when decisions are framed incorrectly as choosing between life and death.[34] Physicians conceded that they struggle to explain illness trajectory and complexity and do not discuss prognosis unless prompted. There are currently extremely limited comparative prognostic data on CKM and dialysis. However, a recent systematic review of survival of CKM versus dialysis for elderly patients demonstrated a broadly similar 1-year survival.[86] Patients who choose CKM tend to have stable symptom burden and HRQOL until the last 1 or 2 months of life, and symptoms are amenable to KSC interventions.[91] Given the degree of uncertainty about benefits and risks and the potentially profound impact on HRQOL, patient values must be paramount in the decision about whether to have dialysis.

There has been great interest in the development of patient decision aids to help with kidney replacement therapy decisions, including CKM. Patient decision aids are tools designed to facilitate people's ability to make reasoned decisions about care, which are compatible with their goals, values, and prognosis. These tools help supplement and guide the shared decision-making process and can be developed in multiple media forms, such as written materials, audiotapes, videos, Web-based formats, and oral presentations. Two Cochrane reviews evaluating the use of patient decision aids have shown them to be effective[97,98] and can improve peoples' knowledge regarding options, reduce their decisional conflict, improve understanding of the relative risks and benefits, increase their participation in decision making, and improve values-choice concordance.[99] However, most decision aids for dialysis treatment options either do not mention CKM or address it only in a peripheral way. These decision aids also do not include patient-specific risks and benefits, so they are not positioned to weigh HRQOL versus survival advantage, which patients require to make the decision that is best for them. New CKM decision aids are currently under development and evaluation to address these deficits.[77,100] These tools, however, must be used with caution. They are educational tools only and should not be used to mandate certain care pathways over others.

PALLIATIVE DIALYSIS

As disease progresses, patients' goals of care may shift to focus almost exclusively on HRQOL rather than survival,

Table 61.7 Summary of Chronic Kidney Disease Management Recommendations for People Receiving Conservative Kidney Management (CKM)[77,78]

Guideline	Treatment Rationale	Recommended Interventions
Dyslipidemia	Patients are unlikely to benefit from treating dyslipidemia in their last few years of life. Patients may gain improvement in HRQOL from stopping statin medications	Care providers, in discussion with the patient, can discontinue statin medications
Blood pressure	The primary goal of blood pressure management is to optimize physical and cognitive function and minimize the risk of falls, while avoiding high readings. Decisions about specific medications would depend on the patient's comorbidities Diuretics are a unique consideration and are aimed primarily at the treatment of volume overload that causes breathlessness or symptomatic peripheral edema	Blood pressure targets can be relaxed for most CKM patients to ≤160/90 mm Hg. This applies to patients with diabetes as well
Sodium restriction	High sodium intake may be associated with volume overload leading to breathlessness and symptomatic peripheral edema Because sodium restriction can influence palatability, strategies should balance symptom management and the patient's priorities for care and other issues such as appropriate nutrition	Diuretics and dietary sodium restriction to assist with volume control if volume overload is contributing to symptom burden
Anemia	Anemia can contribute to fatigue and breathlessness. The purpose of treating anemia would be to reduce these symptoms as opposed to reducing cardiac mortality or morbidity. When a patient starts spending most of his or her time lying down and/or the end of life is near, it is no longer appropriate to manage fatigue and breathlessness by addressing anemia and anemia treatment can be stopped	Erythropoiesis-stimulating agents and iron supplementation Anemia-related bloodwork every 3-6 months is appropriate but should be based on patient preference and symptoms It is appropriate to stop managing anemia in the last weeks or days of life
Hyperkalemia	Hyperkalemia predisposes patients to cardiac arrhythmias and sudden death. Acute treatment of hyperkalemia is appropriate if consistent with the patient's goals If patients wish to liberalize their potassium intake, the risks of lifting the restriction must be explained clearly	Interventions include a potassium-restricted diet and use of potassium-binding resins such as sodium polystyrene sulfonate For those aiming to maintain normal potassium levels, it is reasonable to monitor potassium levels monthly. It is appropriate to stop monitoring and managing potassium levels in the last weeks or days of life.
Acidosis	Metabolic acidosis may contribute to fatigue, bone loss, and muscle wasting; Treatment is aimed at managing these symptoms. If the patient finds the pill burden too great, or when the patient can no longer swallow, treatment should be stopped	The primary intervention is use of sodium bicarbonate. It is reasonable to monitor bicarbonate every 3-6 months if patients are being treated It is appropriate to stop monitoring and managing bicarbonate levels in the last weeks or days of life
Calcium/ Phosphorus	We recommend promoting HRQOL through liberalizing diet and maintaining adequate nutrition rather than normalizing biochemistry as it is not clear that there are benefits to normalizing in people receiving CKM in the last few years of life. Rather, there is a possibility of harm in promoting lower protein intake in patients already at high risk for protein malnutrition. Hyperphosphatemia may contribute to RLS; calcium and phosphorous depositions can lead to myalgias, arthralgias, and pseudogout Interventions to partially normalize biochemistry should only be considered to minimize these symptoms	If the patient is experiencing symptoms, interventions include a phosphorus-restricted diet (being careful to maintain adequate nutrition) and the use of phosphate binders such as calcium carbonate Bloodwork every 3-6 months is appropriate if patients are being treated but should be based on patient preference and symptoms
Vitamin D	Vitamin D may have a role in the symptoms of fatigue, weakness, and muscle loss	Low-dose active vitamin D (vitamin D analog) may be beneficial. There is no additional benefit in monitoring parathyroid hormone levels

HRQOL, Health-related quality of life; *RLS,* restless legs syndrome.

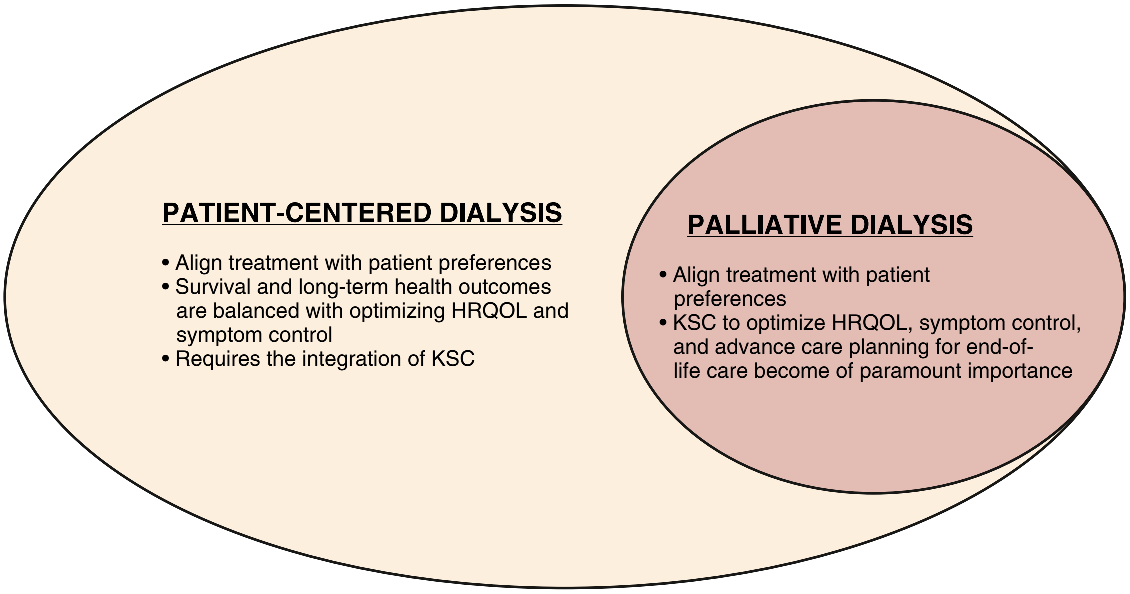

Fig. 61.7 Conceptualizing palliative dialysis. *HRQOL,* Health-related quality of life; *KSC,* kidney supportive care. (Adapted from Davison SN, Jassal SV. Supportive care: integration of patient-centered kidney care to manage symptoms and geriatric syndromes. *Clin J Am Soc Nephrol*. 2016;11:1882–1891.)

with a strong emphasis on symptom control, advance care planning, and emotional, social, and family support. Palliative dialysis is an approach to dialysis that "prioritizes" HRQOL over survival and requires full integration of KSC (Fig. 61.7). Palliative dialysis is often, incorrectly, perceived as being equivalent to less dialysis or a precursor to withdrawal of dialysis. Although adaptation of dialysis timings or eventual withdrawal from dialysis may be a component, this alone will rarely ameliorate symptoms or suffering for patients.[22] This care pathway has recently been explored in greater detail elsewhere.[22] Dialysis patients may transition to a palliative approach to their dialysis care near the end of their life (see Chapter 60). Patients also may start dialysis with this palliative approach to their care. The dialysis guidelines and standards of care will need to permit flexibility so that care can be tailored appropriately for each patient and will look more like the CKM guidelines in Fig. 61.6.

WITHDRAWAL OF DIALYSIS

Withdrawal from dialysis remains a leading cause of death for people on dialysis, although there are significant differences in reported rates across studies and national renal registries[101] and in practice patterns between countries.[102] Likely, many of these differences relate to variations in culture, policy, and physician perceptions and attitudes toward withdrawal. For most cultures, withdrawal of dialysis is ethically and clinically acceptable when based on shared decision making that balances the patient's interests with local cultural norms.[1]

Potential situations in which it is appropriate to withdraw dialysis are described in Box 61.1. It is incumbent upon all providers caring for a patient contemplating stopping dialysis to address potentially remediable factors contributing to the decision, such as depression or other symptoms such as pain, as well as potentially reversible social factors (discussed further in Chapter 60). Ensuring access to appropriate KSC is an integral part of the care following a decision to withdraw dialysis.

Box 61.1 Situations in Which it is Appropriate to Consider Withdrawal From Dialysis[1,20]

1. Patients with decision-making capacity who, being fully informed and making voluntary choices, refuse dialysis or request that dialysis be discontinued.
2. Patients who no longer possess decision-making capacity who have previously indicated refusal of dialysis through appropriate advance care planning.
3. Patients who no longer possess decision-making capacity and whose properly appointed legal agents/surrogates refuse dialysis or request that it be discontinued.
4. Patients with irreversible, profound neurologic impairment such that they lack signs of thought, sensation, purposeful behavior, and awareness of self and environment.

EARLY IDENTIFICATION OF PATIENTS MOST LIKELY TO BENEFIT FROM KIDNEY SUPPORTIVE CARE

A systematic approach is required to identify patients most likely to benefit from KSC. People need to be identified on the basis of need, not just poor prognosis. Not all people with advanced CKD require ongoing KSC, and requirements may fluctuate over the course of their illness trajectory (see Fig. 61.1). However, most people transition to a trajectory of progressive functional decline and/or experience complex clusters of physical and psychological symptoms. Early recognition of needs helps with planning; results in fewer hospital admissions and crises as patients approach the end of life; avoids needless suffering; and ultimately improves both patient and health system outcomes. Once the key elements of KSC are understood, it becomes clearer how to screen for patients in need. Fig. 61.2 highlights key issues that should be considered when screening patients for KSC needs. At a minimum, this would include patients at a high risk of early death, those whose condition is deteriorating, patients with a high symptom burden or poor HRQOL, and those who are struggling with decision making and goals of care.

PEDIATRIC CONSIDERATIONS

Children and young people with CKD experience significant physical, psychological, and existential distress that affects their entire family. Unlike adult nephrology, there has been limited integration of KSC into pediatric nephrology. Special considerations must be made for these children and young people as outlined elsewhere.[103,104] At a minimum, shared decision making should be prioritized and must incorporate the child, parent, and kidney care team. It requires open, honest, age, and developmentally appropriate communication that seeks to understand and respect the child's preferences for involvement and the degree and timing of disclosure.[103]

Children and young people experience high symptom burden including pain, which is often underrecognized.[104] Unfortunately, symptom management has not been well studied in children with advanced CKD and recommendations are based on clinical experience and the adult literature. A holistic approach to symptom management that explores the child's understanding, fears, and wishes for nonpharmacologic strategies such as music or art therapy (multimodal therapy) is essential.[103] Opioid use in children with advanced CKD has been poorly studied. However, there is some evidence that recommendations for the choice of opioids in adults may not be the safest or most effective for children. Rather, morphine and oxycodone are generally the opioids of choice for children with advanced CKD.[103] Integrating holistic KSC into pediatric nephrology for all children and young adults with advanced CKD, from the time of diagnosis, is essential to mitigate suffering and promote quality of life, but further study is desperately needed to advance this field.

SUMMARY

Addressing KSC needs is a high priority for people with advanced CKD. The international kidney community recognizes that access to KSC should be available to all people with advanced CKD, although the extent and form of these services will depend on local resources. For optimal delivery, multidisciplinary teams will need to work collaboratively across health care sectors. There will also need to be an increased emphasis on education in KSC that recognizes this as a core competency in nephrology. Integrating KSC will also require recognition that our current models of care, policies, and clinical standards do not consistently meet the needs of our patients. Ongoing work will help determine best practices, including appropriate quality of KSC metrics.

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QUESTIONS

1. A 62-year-old woman on hemodialysis presents with a 6-month history of a constant dull 6/10 pain in her shoulders and knees. What medication would you use to control her pain initially while investigating the cause of her pain?
   a. Gabapentin 100 mg nightly
   b. Acetaminophen 500–1000 mg three times daily
   c. Nonsteroidal antiinflammatory at a low dose two times daily
   d. Hydromorphone 0.5–1.0 mg every 4–6 hours as needed
   e. No analgesic should be prescribed until the cause of pain is identified and reversible factors addressed

2. A 75-year-old man with an estimated glomerular filtration rate of 9 mL/min/1.73 m^2 being managed with CKM has severe chronic pain from inoperable small vessel peripheral vascular disease that he rates as 7/10. What medications would you use to initiate treatment for his pain?
   a. Gabapentin 100 mg nightly
   b. Acetaminophen 1000 mg three times daily
   c. Gabapentin 100 mg nightly + acetaminophen 1000 mg three times daily
   d. Hydromorphone 0.5–1.0 mg every 4–6 hours as needed
   e. Gabapentin 100 mg nightly + hydromorphone 0.5–1.0 mg every 4–6 hours as needed

3. For whom might CKM be a viable treatment option?
   a. An 84-year-old who is functionally independent and has a long-standing hypertension
   b. A 67-year-old with multiple comorbidities including poor cognitive function
   c. A 45-year-old diabetic with severe vascular disease who is not a transplant candidate and who does not wish to proceed with dialysis
   d. None of the above
   e. b and c

ANSWERS

1. b
Rationale: This woman has nociceptive pain and can be treated with traditional analgesics, rather than an adjuvant such as gabapentin. The first line of therapy is the nonopioid acetaminophen, even though her initial pain is moderate in severity. It is not necessary to wait for a diagnosis to initiate therapy. One should try to initiate analgesia before the pain becomes intolerable and established, as pain is more difficult to treat once the pain cycle becomes established.

2. c
Rationale: Peripheral vascular disease almost always has a neuropathic component, which generally responds poorly to opioids or requires doses for a response that is associated with unacceptable toxicity. Adjuvant analgesics such as gabapentin should be tried first. Given that the initial pain is severe, it is reasonable to add a nonopioid analgesic such as acetaminophen concurrently while the gabapentin is being titrated to maximal effectiveness. Only after the adjuvant analgesic has been titrated to maximal effect should the addition of opioids be considered if the pain remains inadequately controlled. Initiating opioids and gabapentin concurrently in an elderly patient with advanced CKD and multimorbidity would put him at high risk from adverse effects including falls.

3. e
Rationale: Age alone is not an ideal criterion for the consideration of CKM. Elderly patients who remain robust are likely to have a survival advantage, even if it is small, with dialysis. Regardless of age, those who have multiple comorbidities, especially cognitive dysfunction, are not likely to benefit substantially from dialysis and CKM should be considered. Even in younger patients who through informed consent view the burdens of dialysis to outweigh the benefits, CKM is a reasonable option to consider.